Diagnostic questions

Planning questions that reveal understanding

Use diagnostic questions when you need to see which misconception or decision is causing difficulty. A well-designed wrong option tells you something useful about students' thinking.

How to do it well

1 Specify the knowledge. Target one important idea, decision or misconception; minimise unrelated reading and calculation demands.

2 Build the options. Write the correct response and plausible distractors, recording the reasoning each distractor would indicate.

3 Protect independent commitment. Explain the response method first, allow silent thinking and collect everyone's choice before revealing any answer.

4 Probe the pattern. Ask selected students for their reasoning, including some who chose correctly; compare alternatives without shaming individuals.

5 Act on the diagnosis. Reteach the identified distinction, use discussion where students have something useful to compare, and ask a fresh related question.

What this sounds like

"Work out your answer first. Now compare it with the four options. Choose privately and prepare to explain which feature rules out the others."

Classroom example. A Maths teacher designs options from likely errors in calculating a percentage reduction: subtracting the percentage as a number, retaining the original amount, or calculating only the reduction. The distribution reveals which decision needs teaching. Students explain a wrong option before completing a fresh calculation, rather than simply being told which letter is correct.

Top tips that make the technique work

Make distractors diagnostic. Each wrong option should arise from a recognisable error you could respond to.

Avoid accidental clues. Option length, grammar or an implausible answer should not give away the correct choice.

Generate before recognising. Where appropriate, hide the options first so students attempt an answer before selecting.

Investigate wrong options. Ask why a student might choose them, or how to alter an option to make it correct.

Inspect correct reasoning too. A correct choice can come from guessing or a faulty method; probe selected explanations.

Do and avoid

Do	Avoid
Use one unambiguously correct answer.	Writing arbitrary wrong options to fill the spaces.
Interpret the pattern of distractors before deciding the response.	Treating the percentage correct as the whole diagnosis.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton - Diagnostic questions | DistillED - Hinge Questions